

MERN stack - Shopping Online

Lab 03

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1 MERN stack

1.1 Functionals

1.1.1 Admin - listcategory

1.1.1.1 Server

- Create 'server/models/CategoryDAO.js' file

```
1 require('../utils/MongooseUtil');
2 const Models = require('../Models');
3
4 const CategoryDAO = {
5   async selectAll() {
6     const query = {};
7     const categories = await Models.Category.find(query).exec();
8     return categories;
9   }
10 };
11 module.exports = CategoryDAO;
```

- Update 'server/api/admin.js' file

```
1 ...
2 // daos
3 const CategoryDAO = require('../models/CategoryDAO');
4 ...
5 // category
6 router.get('/categories', JwtUtil.checkToken, async function (req, res) {
7   const categories = await CategoryDAO.selectAll();
8   res.json(categories);
9 });
10 ...
```

- Postman: (GET) http://localhost:3000/api/admin/categories

- Headers: "x-access-token": <token>

1.1.1.2 Client-admin

- Update 'client-admin/src/components/MainComponent.js' file

```
1 ...
2 import Category from './CategoryComponent';
3
4 class Main extends Component {
5   ...
6   render() {
7     ...
8     return (
9       ...
10      <Routes>
11        ...
12        <Route path='/admin/category' element={<Category />} />
13      </Routes>
14    );
15    ...
16  }
17 }
18 ...
```

- Update 'client-admin/src/components/MenuComponent.js' file

```
1 ...
2 class Menu extends Component {
3   ...
4   render() {
5     return (
```

```

6      ...
7      <ul className="menu">
8          ...
9          <li className="menu"><Link to='/admin/category'>Category</Link></li>
10         ...
11     </ul>
12     ...
13 );
14 }
15 ...
16 }
17 ...

```

- Create 'client-admin/src/components/CategoryComponent.js' file

```

1  import axios from 'axios';
2  import React, { Component } from 'react';
3  import MyContext from '../contexts/MyContext';
4  import CategoryDetail from './CategoryDetailComponent';
5
6  class Category extends Component {
7      static contextType = MyContext; // using this.context to access global state
8      constructor(props) {
9          super(props);
10         this.state = {
11             categories: [],
12             itemSelected: null
13         };
14     }
15     render() {
16         const cates = this.state.categories.map((item) => {
17             return (
18                 <tr key={item._id} className="datatable" onClick={() => this.trItemClick(item)}>
19                     <td>{item._id}</td>
20                     <td>{item.name}</td>
21                 </tr>
22             );
23         });
24         return (
25             <div>
26                 <div className="float-left">
27                     <h2 className="text-center">CATEGORY LIST</h2>
28                     <table className="datatable" border="1">
29                         <tbody>
30                             <tr className="datatable">
31                                 <th>ID</th>
32                                 <th>Name</th>
33                             </tr>
34                             {cates}
35                         </tbody>
36                     </table>
37                 </div>
38                 <div className="inline" />
39                 <CategoryDetail item={this.state.itemSelected} />
40                 <div className="float-clear" />
41             </div>
42         );
43     }
44     componentDidMount() {
45         this.apiGetCategories();
46     }
47     // event-handlers
48     trItemClick(item) {
49         this.setState({ itemSelected: item });
50     }
51     // apis
52     apiGetCategories() {
53         const config = { headers: { 'x-access-token': this.context.token } };
54         axios.get('/api/admin/categories', config).then((res) => {

```

```

55     const result = res.data;
56     this.setState({ categories: result });
57   });
58 }
59 }
60 export default Category;

```

- Create 'client-admin/src/components/CategoryDetailComponent.js' file

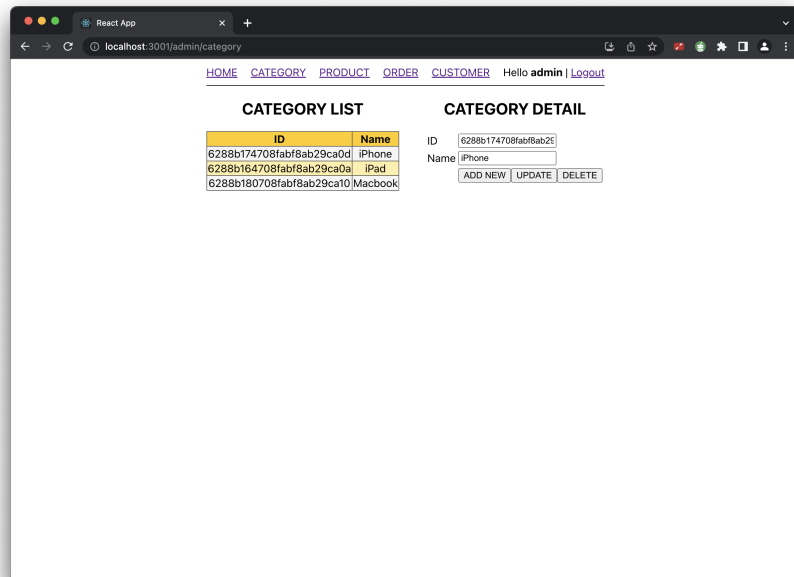
```

1  import axios from 'axios';
2  import React, { Component } from 'react';
3  import MyContext from '../contexts/MyContext';
4
5  class CategoryDetail extends Component {
6    static contextType = MyContext; // using this.context to access global state
7    constructor(props) {
8      super(props);
9      this.state = {
10        txtID: '',
11        txtName: ''
12      };
13    }
14    render() {
15      return (
16        <div className="float-right">
17          <h2 className="text-center">CATEGORY DETAIL</h2>
18          <form>
19            <table>
20              <tbody>
21                <tr>
22                  <td>ID</td>
23                  <td><input type="text" value={this.state.txtID} onChange={(e) => { this.
24                      setState({ txtID: e.target.value }) }} readOnly={true} /></td>
25                </tr>
26                <tr>
27                  <td>Name</td>
28                  <td><input type="text" value={this.state.txtName} onChange={(e) => { this.
29                      setState({ txtName: e.target.value }) }} /></td>
30                </tr>
31                <tr>
32                  <td></td>
33                  <td>
34                    <input type="submit" value="ADD NEW" />
35                    <input type="submit" value="UPDATE" />
36                    <input type="submit" value="DELETE" />
37                  </td>
38                </tr>
39              </tbody>
40            </table>
41          </form>
42        </div>
43      );
44    }
45    componentDidUpdate(prevProps) {
46      if (this.props.item !== prevProps.item) {
47        this.setState({ txtID: this.props.item._id, txtName: this.props.item.name });
48      }
49    }
50  }
51  export default CategoryDetail;

```

- Browser: <http://localhost:3001/admin/category>

1.1.1.3 Screenshots



1.1.2 Admin - addcategory

1.1.2.1 Server

- Update 'server/models/CategoryDAO.js' file

```

1  ...
2  const CategoryDAO = {
3    ...,
4    async insert(category) {
5      const mongoose = require('mongoose');
6      category._id = new mongoose.Types.ObjectId();
7      const result = await Models.Category.create(category);
8      return result;
9    }
10 };
11 ...

```

- Update 'server/api/admin.js' file

```

1  ...
2  // category
3  router.post('/categories', JwtUtil.checkToken, async function (req, res) {
4    const name = req.body.name;
5    const category = { name: name };
6    const result = await CategoryDAO.insert(category);
7    res.json(result);
8  });
9  ...

```

- Postman: (POST) http://localhost:3000/api/admin/categories

- Headers: "x-access-token": <token>
- Body (raw+JSON): { "name": "Watch" }

1.1.2.2 Client-admin

- Update 'client-admin/src/components/CategoryComponent.js' file

```

1  ...
2  class Category extends Component {
3    ...
4    render() {
5      ...

```

```

6     return (
7         <div>
8             ...
9             <CategoryDetail item={this.state.item} updateCategories={this.updateCategories} />
10            ...
11        </div>
12    );
13 }
14 updateCategories = (categories) => { // arrow-function
15     this.setState({ categories: categories });
16 }
17 ...
18 }
19 ...

```

- Update 'client-admin/src/components/CategoryDetailComponent.js' file

```

1 ...
2 class CategoryDetail extends Component {
3     ...
4     render() {
5         ...
6         return (
7             ...
8             <input type="submit" value="ADD NEW" onClick={(e) => this.btnAddClick(e)} />
9             ...
10        );
11        ...
12    }
13    // event-handlers
14    btnAddClick(e) {
15        e.preventDefault();
16        const name = this.state.txtName;
17        if (name) {
18            const cate = { name: name };
19            this.apiPostCategory(cate);
20        } else {
21            alert('Please input name');
22        }
23    }
24    // apis
25    apiPostCategory(cate) {
26        const config = { headers: { 'x-access-token': this.context.token } };
27        axios.post('/api/admin/categories', cate, config).then((res) => {
28            const result = res.data;
29            if (result) {
30                alert('OK BABY!');
31                this.apiGetCategories();
32            } else {
33                alert('SORRY BABY!');
34            }
35        });
36    }
37    apiGetCategories() {
38        const config = { headers: { 'x-access-token': this.context.token } };
39        axios.get('/api/admin/categories', config).then((res) => {
40            const result = res.data;
41            this.props.updateCategories(result);
42        });
43    }
44    ...
45 }
46 ...

```

- Browser: <http://localhost:3001/admin/category>

1.1.3 Admin - updatecategory

1.1.3.1 Server

- Update 'server/models/CategoryDAO.js' file

```
1 ...
2 const CategoryDAO = {
3   ...,
4   async update(category) {
5     const newvalues = { name: category.name }
6     const result = await Models.Category.findByIdAndUpdate(category._id, newvalues, { new: true
7     });
8     return result;
9   }
10  };
11  ...
```

- Update 'server/api/admin.js' file

```
1 ...
2 // category
3 router.put('/categories/:id', JwtUtil.checkToken, async function (req, res) {
4   const _id = req.params.id;
5   const name = req.body.name;
6   const category = { _id: _id, name: name };
7   const result = await CategoryDAO.update(category);
8   res.json(result);
9 });
10 ...
```

- Postman: (PUT) <http://localhost:3000/api/admin/categories/<id>>

- Headers: "x-access-token": <token>
- Body (raw+JSON): { "name": "Watch2" }

1.1.3.2 Client-admin

- Update 'client-admin/src/components/CategoryDetailComponent.js' file

```
1 ...
2 class CategoryDetail extends Component {
3   ...
4   render() {
5     ...
6     return (
7       ...
8       <input type="submit" value="UPDATE" onClick={(e) => this.btnUpdateClick(e)} />
9       ...
10    );
11    ...
12  }
13  // event-handlers
14  btnUpdateClick(e) {
15    e.preventDefault();
16    const id = this.state.txtID;
17    const name = this.state.txtName;
18    if (id && name) {
19      const cate = { name: name };
20      this.apiPutCategory(id, cate);
21    } else {
22      alert('Please input id and name');
23    }
24  }
25  // apis
26  apiPutCategory(id, cate) {
27    const config = { headers: { 'x-access-token': this.context.token } };
28    axios.put('/api/admin/categories/' + id, cate, config).then((res) => {
29      const result = res.data;
30      if (result) {
31        alert('OK BABY!');
32        this.apiGetCategories();
33      }
34    });
35  }
36}
```

```

33     } else {
34       alert('SORRY BABY!');
35     }
36   });
37 }
38 ...
39 }
40 ...

```

- Browser: <http://localhost:3001/admin/category>

1.1.4 Admin - deletecategory

1.1.4.1 Server

- Update 'server/models/CategoryDAO.js' file

```

1 ...
2 const CategoryDAO = {
3   ...,
4   async delete(_id) {
5     const result = await Models.Category.findByIdAndRemove(_id);
6     return result;
7   }
8 };
9 ...

```

- Update 'server/api/admin.js' file

```

1 ...
2 // category
3 router.delete('/categories/:id', JwtUtil.checkToken, async function (req, res) {
4   const _id = req.params.id;
5   const result = await CategoryDAO.delete(_id);
6   res.json(result);
7 });
8 ...

```

- Postman: (DELETE) <http://localhost:3000/api/admin/categories/<id>>

- Headers: "x-access-token": <token>

1.1.4.2 Client-admin

- Update 'client-admin/src/components/CategoryDetailComponent.js' file

```

1 ...
2 class CategoryDetail extends Component {
3   ...
4   render() {
5     ...
6     return (
7       ...
8       <input type="submit" value="DELETE" onClick={(e) => this.btnDeleteClick(e)} />
9       ...
10    );
11    ...
12  }
13  // event-handlers
14  btnDeleteClick(e) {
15    e.preventDefault();
16    if (window.confirm('ARE YOU SURE?')) {
17      const id = this.state.txtID;
18      if (id) {
19        this.apiDeleteCategory(id);
20      } else {
21        alert('Please input id');
22      }

```

```
23     }
24   }
25   // apis
26   apiDeleteCategory(id) {
27     const config = { headers: { 'x-access-token': this.context.token } };
28     axios.delete('/api/admin/categories/' + id, config).then((res) => {
29       const result = res.data;
30       if (result) {
31         alert('OK BABY!');
32         this.apiGetCategories();
33       } else {
34         alert('SORRY BABY!');
35       }
36     });
37   }
38   ...
39 }
40 ...
```

- Browser: <http://localhost:3001/admin/category>